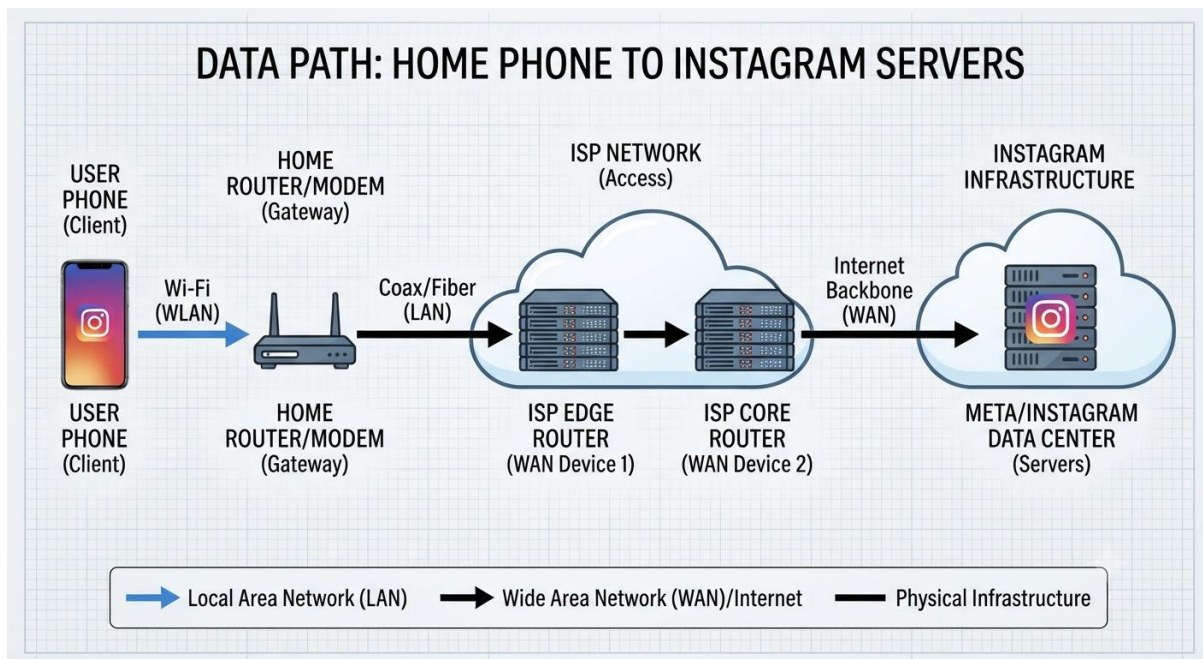


Day – 5 Wide Area Networks.

Task – 1 Diagram client to server.



Task – 2 Types of WAN technologies.

- 1. Broadband Internet
- Speed & Performance: Offers asymmetrical speeds (faster downloads, slower uploads), shared bandwidth among multiple users, and variable performance during peak hours.
- Cost: Low cost with affordable monthly subscriptions and minimal setup fees.
- Real-Life Use Case: Your Home – used for streaming movies, attending online classes, or browsing social media.
- 2. Leased Line (Internet Leased Line / ILL)
- Speed & Performance: Provides high, fully dedicated, symmetrical speeds (equal upload and download rates) backed by strict uptime SLAs and low latency.
- Cost: High cost, involving expensive installation and premium monthly recurring charges.
- Real-Life Use Case: IRCTC Ticketing Center – handling massive, concurrent high-volume booking transactions without downtime or performance drops.

- 3. MPLS (Multiprotocol Label Switching)
- Speed & Performance: Features high-performance routing with low latency, label-based packet routing, and built-in traffic prioritization (Quality of Service / QoS) across private paths.
- Cost: Moderate to high. It is cost-effective for multi-site enterprise networking compared to deploying multiple individual point-to-point leased lines, but more expensive than standard broadband.
- Real-Life Use Case: Zomato Corporate Network – securely connecting regional offices, branch locations, and backend data centers to streamline internal operations.

Task – 3 How VPN works.

- 1. Establishing the Encrypted Tunnel
- When you turn on a VPN app on your phone and select a server, your device initiates a secure handshake with that remote VPN server. This process creates a virtual "tunnel" through the public WAN infrastructure (the coffee shop's local router, the local ISP, and the internet backbone).
- 2. Encapsulating Your App Traffic
- Whenever you stream a song on Spotify or watch a video on YouTube, your phone generates data packets. Normally, these packets travel across the public WAN in plain text or standard encryption that local network administrators or hackers can monitor. With a VPN active, your device encapsulates these packets—wrapping them inside an extra layer of highly secure encryption before they leave your phone.
- 3. Traversing the Public WAN Safely
- As your encapsulated data packets travel across the public WAN routers and switches, anyone attempting to intercept them (such as a malicious actor on the same coffee shop Wi-Fi) will only see scrambled, unreadable ciphertext. To the public network, your traffic simply looks like a stream of secure, indecipherable data heading toward the VPN server.
- 4. Decryption and Destination Exit
- Once the encrypted packets reach the remote VPN server, the server strips away the outer encryption layer, decrypts your original request, and safely forwards it to Spotify's or YouTube's servers. When the music or video data returns, the VPN server encrypts it again, sends it back through the private tunnel to your phone, where your app decrypts and plays it.

Task – 4 Based on scenario.

- 1. High Security and Private Infrastructure
- Warehouses constantly transmit sensitive operational data, including inventory logs, real-time stock updates, financial transactions, and customer shipping details. Unlike standard broadband, which traverses the public internet, MPLS operates over a private, dedicated routing backbone. This prevents data from being exposed to public internet vulnerabilities, significantly lowering the risk of interception or cyberattacks during inter-city data transfers.
- 2. Guaranteed Quality of Service (QoS) and Low Latency
- Flipkart's fulfillment centers rely heavily on instantaneous barcode scanning, automated sorting systems, and synchronized inventory databases where even a split-second delay can cause bottlenecks or order fulfillment errors. MPLS allows network administrators to prioritize critical traffic using built-in Quality of Service (QoS) labels, ensuring consistent, low-latency bandwidth specifically reserved for business-critical applications regardless of public network congestion.

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